Economic growth and human well-being

Does a country's income tell us how well its people live? 148 countries, 2000–2022. Luyu Su & Yifan Zuo · STATS 401 · yyiffann.github.io/stats401-final-project

WHY AND QUESTIONS

GDP per person is the default yardstick of progress, but says little about health, schooling, services, work and clean air.

Key measure: the **well-being gap** = well-being score minus the score expected at that income.

- Where is prosperity, and where does well-being beat it?
- How did countries' paths differ?
- Who departs from the income pattern?
- Which dimensions explain it?
- Who moved to a higher stage?

DATA AND METHOD

- World Bank WDI + UNDP HDR + World Happiness Report + Natural Earth
- 148 countries × 23 years = 3,404 rows; 10 indicators → 5 dimensions → 1 score (fixed goalposts)
- Yearly LOWESS of score on log GDP gives the gap
- 2,000 random weightings test robustness
- k-means (k=5) development profiles at 4 snapshot years
- Validation: score vs. life satisfaction r = 0.78
- D3.js v7, one shared state links all views

KEY FINDINGS

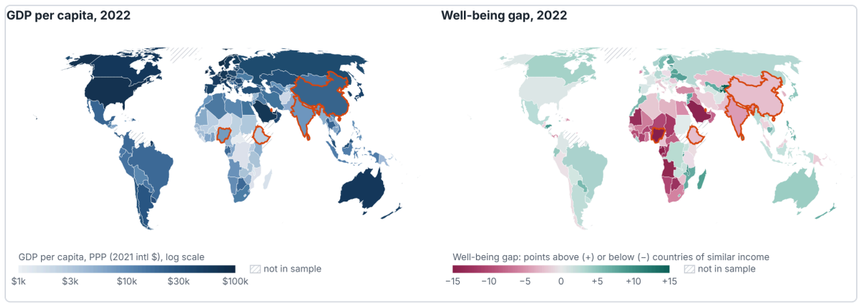
+16 → −21
gap range 2022, Kyrgyz Republic to Qatar

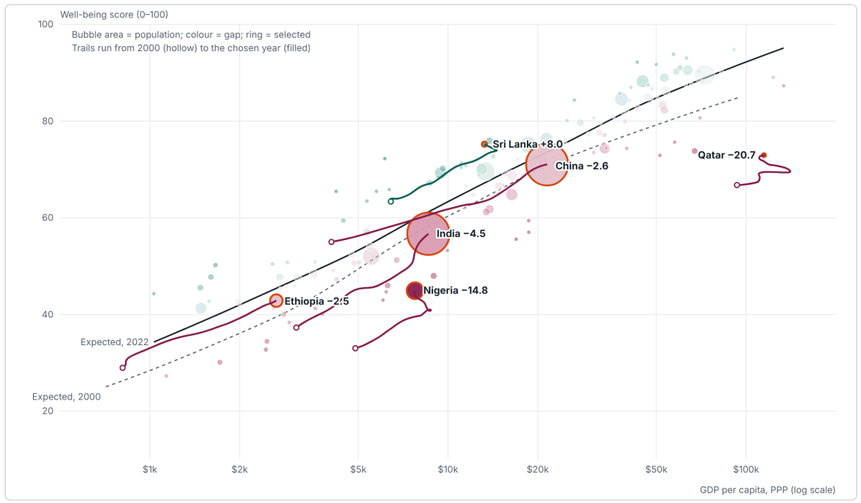
36 → 42
expected score at $2,000, 2000 → 2022

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"catching up" countries reached upper-middle

35% → 13%
people in the lowest profile, 2000 → 2022

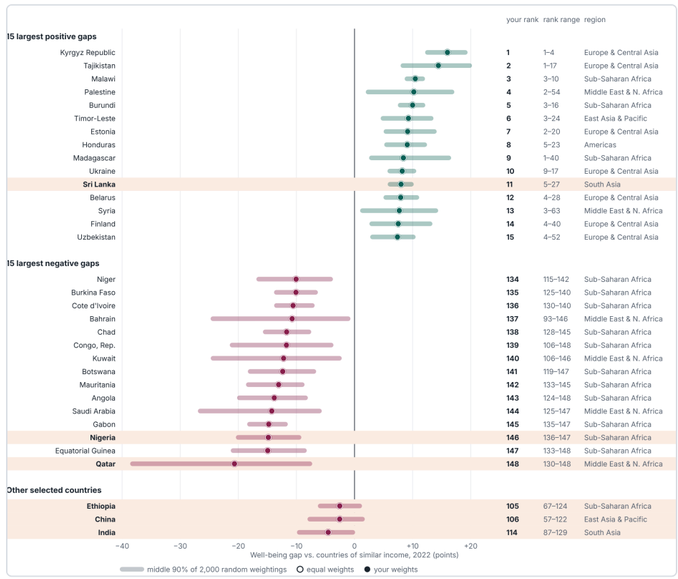
1 · WHERE (MAP) & 2 · HOW (PATHS)

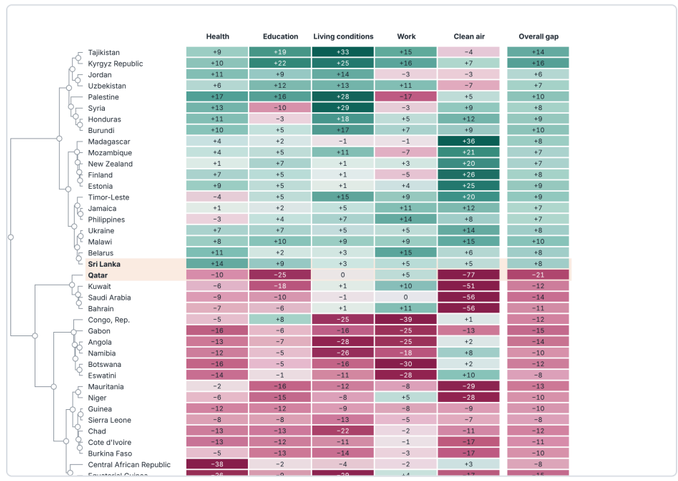




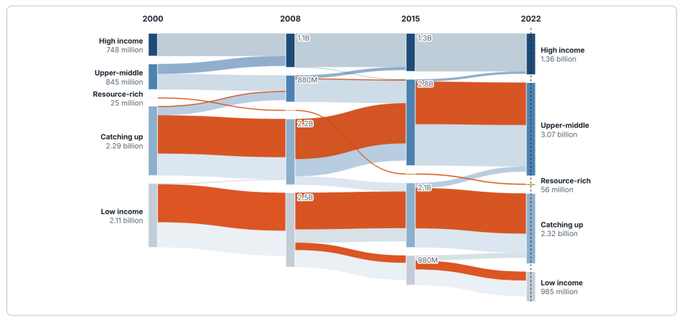
Year slider animates both; brush on the plot selects groups; selection is shared across all views.

3 · WHO (WEIGHTS) & 4 · WHY (HEATMAP)





5 · WHO MOVED (ALLUVIAL)



Gulf states lose on clean air and education; African resource exporters on health, living conditions and work.